Introduction to Machine Learning and Optimization

Supervised learning is a machine learning paradigm where models are trained on labeled data.

Common supervised learning algorithms include linear regression, logistic regression, support vector m

and neural networks. The objective is to minimize a loss function using optimization techniques like gra

Gradient descent iteratively updates model weights in the direction opposite to the gradient of the loss

Learning rate is a critical hyperparameter that controls the step size during parameter updates in gradi

Page 2: Overfitting, Regularization, and Model Evaluation

Overfitting occurs when a machine learning model learns the training noise instead of the underlying p

To combat overfitting, practitioners employ regularization techniques such as L1 Lasso and L2 Ridge r

Cross-validation, particularly k-fold cross validation, is essential to estimate model generalization perfo

Evaluating models on unseen test datasets prevents over-optimistic performance metrics and data lea

Unsupervised learning algorithms, such as k-means clustering and PCA, discover hidden structures in